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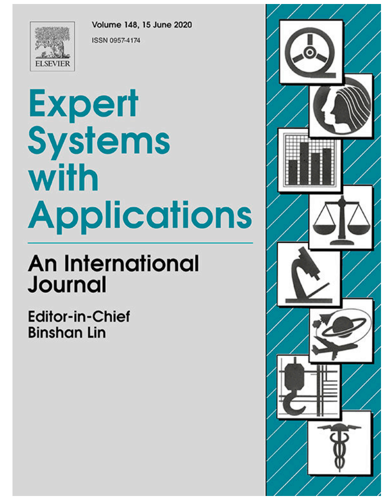

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Deep multi-hybrid forecasting system with random EWT extraction and variational learning rate algorithm for crude oil futures

Bin Wang [1], Jun Wang [1]

School of Science, Beijing Jiaotong University

Beijing 100044, P.R. CHINA

Email: wangbin1@bjtu.edu.cn, wangjun@m.bjtu.edu.cn

Abstract Machine learning algorithms provide feasibility for crude oil price prediction. In this paper, a novel multi-hybrid predictive neural network model is proposed based on complex deep learning algorithm, which integrates empirical wavelet transform, random inheritance formula error correction algorithm, deep bidirectional LSTM neural network and Elman recurrent neural network with variational learning rate. The prediction model is selected according to the sequence frequency after EWT feature extraction, and the prediction results are obtained by separately predicting and reintegrating. On the basis of individual model, the structure of deep bidirectional training, random inheritance formula and variational learning rate are proposed, which further ameliorate the performance of the model and achieve more effective data information capture. Simultaneously, the examination of variational learning rate provides us with a feasible parameter selection. The proposed model achieves high-precision prediction of crude oil futures price, and stands out in the multi-model comparison analysis and q-DSCID synchronous evaluation, with superior prediction accuracy.

Keywords Machine learning; Long short-term memory; Empirical wavelet transform;

[1]Corresponding authors at: Beijing Jiaotong University, Beijing 100044, China. Email: wangbin1@bjtu.edu.cn (B. Wang), wangjun@m.bjtu.edu.cn (J. Wang).

Deep bidirectional training structure; Random inheritance formula; Variational learning rate

Appendix

VR	Variational learning rate
EWT	Empirical wavelet transform
RIF	Random inheritance formula
SVM	Support vector machine
RNN	Recurrent neural network
GRU	Gated recurrent unit
ERNN	Elman recurrent neural network
VRERNN	Elman recurrent neural network with variational learning rate
RIFERNN	Elman recurrent neural network with random inheritance formula
LSTM	Long short-term memory
DBLSTMNN	Deep bidirectional long short-term memory neural networks
RIF-GRUNN	Long short-term memory neural networks with RIF based error correction algorithm
RIF-DBGRUNN	Deep bidirectional long short-term memory neural networks with RIF based error correction algorithm
CID	Complexity invariant distance
q-DSCID	q-order dyadic scales complexity invariant distance
WTI	West Texas Intermediate crude oil
BRE	ICE Brent crude oil
NG	Henry Hub natural gas
HO	Heating oil

1. Introduction

As the "blood" of modern industrial production, crude oil has firmly dominated an inviolable position (Hart, 2016). It is an indispensable and significant resource in industrial field or in life. In recent years, the crude oil supply is unbalanced and the world economic market is unstable (Motlaghi, Jalali, & Ahmadabadi, 2008; Gong & Lin, 2018). Due to the large transmutations in political structure and the frequent occurrence of uncertain events, international crude oil prices have experienced relatively fierce fluctuations, implicating the nerves of countries around the world (Cheng, Li, Wei, & Wei, 2019; Liu, Cheng, Gan, Wang, Li, & Zhao, 2019). Grasping the price of crude oil will better control the lifeline of industry and economy, so forecasting the price of crude oil is of substantial significance (Jammazi & Aloui, 2012; E, Bao, & Ye, 2017).

Artificial neural networks (ANN) provide methods for deciding such nonlinear prediction problems, and have been widely employed in many fields such as language recognition, image recognition and understanding, computer vision, market analysis, decision optimization, expert systems, policy support systems, etc (Riahi-Madvar, Ayyoubzadeh, & Atani, 2011; Yu & Wang, 2012; Li, Chen, Sun, & Tao, 2015; Keles, Scelle, Paraschiv, & Fichtner, 2016; Rizkin, Popovich, & Hartman, 2019). Kristjanpoller and Minutolo (2016) applied the ANN to the GARCH model to construct a new hybrid forecasting model for crude oil price forecasting. The results show that the ANN-GARCH model can successfully improve the volatility forecast and the spot price is higher than the traditional forecasting model. Wang, Dong, and Sun (2010) utilized genetic algorithm to find the optimal network architecture and parameters of artificial neural network for the prediction of acid VGO saturation and achieved good results. Artificial neural networks have good learning ability but also have many shortcomings, such as lack of theoretical derivation, single structure and lack of time correlation between data. The support vector machine (SVM) is a theoretical derivation method, which can obtain the optimal solution but is not suitable for the prediction problem of large sample sequences (Guo, Li, & Zhang, 2012; Kao, Chiu, Lu, & Yang, 2013; Fu, Li, Sun, & Li, 2019; Xu, Wang, Jiang, & Liu, 2019). The recurrent neural network (RNN) is a structural optimization of the ANN, which settles with the problem of correlation between data (Rahman, Srikumar, & Smith, 2018; Hajiabotorabi, Kazemi, Samavati, & Ghaini, 2019; Pan, Zheng, Zhang, & Yao, 2019). Berradia and Lazaara (2019) employed recurrent neural network to predict time series and combines principal component analysis to predict the 29-day total Maroc stock price on the Casablanca Stock Exchange, improving the accuracy of the recursive neural network model and providing good forecasts for stock prices. Chow (2020) used a recurrent neural network combined with deep learning to estimate the license plate auction price. The results show that the accuracy is much higher than other models. With the deepening of the research, the gradient disappearance problem of RNN and the long sequence dependence problem were proposed. In order to further optimize the prediction results, scholars constantly optimized the model structure and proposed long short-term memory (LSTM), which combined with more precise structure to solve the gradient disappearance and long-term dependence problem of RNN (Hochreiter & Schmidhuber, 1997; Kim & Won, 2018). Peng, Liu, Liu, and Wang (2018) introduced effective LSTM for electricity price prediction, and combined with differential evolution algorithm to identify hyperparameters suitable for LSTM. The experimental results show that the proposed model is superior to the existing prediction model in prediction accuracy. Muzaffar and Afshari (2019) applied LSTM to predict electrical load data and compared the predicted results with traditional prediction methods to conclude that LSTM-based predictions are superior to other methods and may further improve prediction accuracy.

The optimization of model structure has been continuously explored to ameliorate the accuracy of model prediction. Simultaneously, scholars have found that feature extraction of initial time series can effectively improve the property of the model. In recent years, the methods of hybrid neural network prediction models combined with feature extraction algorithms have gradually enhanced and become the concentrated issues, such as discrete wavelet decomposition (DWT), empirical mode decomposition (EMD), variational mode decomposition (VMD) and ensemble empirical mode decomposition (EEMD) (Yu, Wang, & Lai, 2008; Wang, Chau, Qiu, & Chen, 2015; Hayat & Qazi, 2017; He, Zhou, Feng, Liu, & Yang, 2019; Mouatadid, Adamowski, Tiwari, & Quilty, 2019). Among them, the decomposition method based on EMD and EEMD is a proverbially employed. Büyükşahin and Ertekin (2019) provided a new hybrid method of autoregressive integrated moving average-ANN hybrid method, combined with EMD technology, and the results show that the hybrid method with EMD can be an effective way to improve forecasting accuracy obtained by traditional hybrid methods and also any of the individual methods. Wu, Wu, and Zhu (2019) proposed an integrated model based on EEMD and LSTM for predicting the spot price of crude oil from West Texas Intermediate (WTI). The empirical results show that the model works well when the number of decomposition results changes, and has broad prospects for predicting crude oil prices. The EMD algorithm is advanced by Huang et al. (1998), which can better achieve the mode decomposition of the signal, but lacks the theoretical foundation and the phenomenon of modal aliasing (Çelebi & Ertürk, 2012; Bisoi, Dash, & Mishra, 2019). The empirical wavelet transform (EWT) algorithm is a novel feature extraction algorithm superior to EMD proposed on the basis of EMD (Gilles, 2013). It precludes mode aliasing by reasonable segmentation and boundary setting, and possess a solid theoretical foundation. However, there are few hybrid prediction methods related to EWT feature extraction.

This paper constructs a novel multi-hybrid predictive neural network model EWT-RIF-DBLSTM-VRERNN. Based on LSTM, combined with deep learning algorithm to establish a bidirectional training network structure to implement efficient extraction of effective data information. Premeditating the dependence of future results on historical data, the random inheritance formula (RIF) is proposed and used to error correction process of model training, which more reasonably reflects the inheritance relationship to historical data (Niu & Wang, 2013). The EWT algorithm is applied to extract overall trend characteristics and local fluctuation characteristics of initial sequences which are predicted by appropriate models. In the actual research, it is found that Elman recurrent neural network (ERNN) has better prediction performance on high frequency sequences. Considering the sensitivity of ERNN to the parameter learning rate, a variational learning rate method is proposed to optimize the learning rate selection of ERNN. The RIF-DBLSTMNN is used to predict the trend feature sequence (low-frequency sequence), and the RIF-VRERNN model is used to predict the local fluctuation feature sequences (high-frequency sequences). Finally, the prediction results of the feature sequences are accumulated to obtain the prediction of the initial sequence.

The proposed model is utilized for crude oil futures price forecasting. In order to estimate the prediction result of the model, SVM, gated recurrent unit (GRU), ERNN, LSTM and many comparison models integrated with the proposed optimization method are compared with the proposed model. Combining error metrics and regression analysis results, the superiority of the proposed model is proved. Simultaneously, a novel q-order dyadic scales complexity invariant distance (q-DSCID) synchronization evaluation algorithm is used in comparative evaluations of the model (Wu, Wu, Lin, Wang, & Lee, 2013; Niu & Wang, 2015).

Contributions of this paper can be summarized as three points. First, a multi-hybrid neural network with EWT extraction is proposed for crude oil futures price forecasting. Second, novel random inheritance formula and variational learning rate are proposed and used for model construction and optimization. Third, comprehensive error evaluation indexes and q-DSCID synchronization error evaluation method are used to compare the proposed model with the existing models. The rest of the paper is organized as follows. Section 2 introduces the proposed model and optimization algorithms. Section 3 describes the basic error metrics and q-DSCID synchronization evaluation arithmetic. Section 4 evaluates the proposed model and compares the superiority of the model. Section 5 is an empirical study of the variational learning rate. Section 6 describes the summary and expectations of the proposed model.

2. Methodology

2.1. Description of empirical wavelet transform

Empirical wavelet transform is a novel signal decomposition algorithm, which was first proposed by (Gilles, 2013). The experimental results show that the EWT is a novel advanced and effective signal analysis algorithm than traditional wavelet decomposition or empirical mode decomposition method. Based on the theory of wavelet, the EWT applies a new adaptive wavelet extraction method. Wavelet analysis theory shows that the initial signal can be resolved into different frequency parts. Because the separated signal is simpler in frequency component than the initial signal, the signal is smoothed by wavelet decomposition. Therefore, it has better theoretical basis, and it is more efficient and meaningful to extract features from financial time series. However, the application of empirical wavelet transform to feature extraction of financial time series is rare. The steps of EWT can be defined as follows. Assume that the Fourier spectrum of the original crude oil futures price sequence is segmented into N consecutive segments. Then, define the empirical scaling function $\hat{\phi}_n(\omega)$ and the empirical wavelets $\hat{\psi}_n(\omega)$ by expressions as follows.

$$\hat{\phi}_n(\omega) = \begin{cases} 1 & |\omega| \le \omega_n - \tau_n \\ \cos\left[\frac{\pi}{2}\beta\left(\frac{1}{2\tau_n}\left(|\omega| - \omega_n + \tau_n\right)\right)\right] & \omega_n - \tau_n \le |\omega| \le \omega_n + \tau_n \\ 0 & \text{otherwise} \end{cases} \tag{1}$$

$$\hat{\psi}_n(\omega) = \begin{cases} 1 & \omega_n + \tau_n \le |\omega| \le \omega_{n+1} - \tau_{n+1} \\ \cos\left[\frac{\pi}{2}\beta\left(\frac{1}{2\tau_{n+1}}\left(|\omega| - \omega_{n+1} + \tau_{n+1}\right)\right)\right] & \omega_{n+1} - \tau_{n+1} \le |\omega| \le \omega_{n+1} + \tau_{n+1} \\ \sin\left[\frac{\pi}{2}\beta\left(\frac{1}{2\tau_n}\left(|\omega| - \omega_n + \tau_n\right)\right)\right] & \omega_n - \tau_n \le |\omega| \le \omega_n + \tau_n \\ 0 & \text{otherwise} \end{cases} \tag{2}$$

where ω_n and τ_n represent the limits between each segments and transition phase respectively. The function $\beta(x)$ is defined as

$$\beta(x) = x^4\left(35 - 84x + 70x^2 - 20x^3\right). \tag{3}$$

Here, we choose τ_n in proportion to ω_n: $\omega_n = \alpha\omega_n$ where $0 < \alpha < n$. Consequently, the equation $\hat{\phi}_n(\omega)$ and $\hat{\psi}_n(\omega)$ can be expressed as

$$\hat{\phi}_n(\omega) = \begin{cases} 1 & |\omega| \leqslant (1-\alpha)\omega_n \\ \cos\left(\frac{\pi}{2}\beta\left(\frac{1}{2\alpha\omega_n}\left(|\omega| - (1-\alpha)\omega_n\right)\right)\right) & (1-\alpha)\omega_n \leqslant |\omega| \leqslant (1+\alpha)\omega_n \\ 0 & \text{otherwise} \end{cases} \tag{4}$$

$$\hat{\psi}_n(\omega) = \begin{cases} 1 & (1+\alpha)\omega_n \leqslant |\omega| \leqslant (1-\alpha)\omega_{n+1} \\ \cos\left(\frac{\pi}{2}\beta\left(\frac{1}{2\alpha\omega_{n+1}}\left(|\omega| - (1-\alpha)\omega_{n+1}\right)\right)\right) & (1-\alpha)\omega_{n+1} \leqslant |\omega| \leqslant (1+\alpha)\omega_{n+1} \\ \sin\left(\frac{\pi}{2}\beta\left(\frac{1}{2\alpha\omega_n}\left(|\omega| - (1-\alpha)\omega_n\right)\right)\right) & (1-\alpha)\omega_n \leqslant |\omega| \leqslant (1+\alpha)\omega_n \\ 0 & \text{otherwise} \end{cases}. \tag{5}$$

To obtain a tight frame, limit $\alpha < \min_n \left(\frac{\omega_{n+1}-\omega_n}{\omega_{n+1}+\omega_n}\right)$. The detail coefficient is expressed as

$$\mathcal{W}_f^{\mathcal{E}}(n,t) = \langle f, \psi_n \rangle = \left(\widehat{f}(\omega)\overline{\widehat{\psi}_n(\omega)}\right)^{\vee}. \tag{6}$$

The approximation coefficient is expressed as

$$\mathcal{W}_f^{\mathcal{E}}(0,t) = \langle f, \phi_1 \rangle = \left(\widehat{f}(\omega)\overline{\widehat{\phi}_1(\omega)}\right)^{\vee}. \tag{7}$$

The reconstruction function is obtained by

$$f(t) = \left(\widehat{\mathcal{W}_f^{\varepsilon}}(0,\omega)\hat{\phi}_1(\omega) + \sum_{n=1}^{N} \widehat{\mathcal{W}_f^{\varepsilon}}(n,\omega)\hat{\psi}_n(\omega)\right)^{\vee}. \tag{8}$$

The empirical mode f_k can be obtained by

$$f_0(t) = \mathcal{W}_f^{\mathcal{E}}(0,t) * \phi_1(t), \quad f_k(t) = \mathcal{W}_f^{\mathcal{E}}(k,t) * \psi_k(t). \tag{9}$$

A more detailed introduction to EWT, including the construction of related functions and parameter selection, can be found in (Gilles, 2013).

2.2. Long short-term memory

LSTM is a recurrent neural network with better structure and more excellent results. It solves the problem of long-term and short-term dependence of traditional neural networks, and avoids the problem of gradient disappearance and gradient explosion during training. Numerous advantages make LSTM one of the most popular neural network models. Its structure unit is shown in Fig. 1. Its structure can be summarized as three-door two-state, forgetting gate, input gate, output, and implicit state, cell state. The forget gate eliminates the information to be forgotten, the input gate determines the information saved in the status cell, and the output gate calculates the output information. The forward propagation process of LSTM is defined as

$$\begin{cases} f_t = \sigma\left(W_f \cdot [h_{t-1}, x_t] + b_f\right) \\ i_t = \sigma\left(W_i \cdot [h_{t-1}, x_t] + b_i\right) \\ \tilde{c}_t = \tanh\left(W_c \cdot [h_{t-1}, x_t] + b_c\right) \\ c_t = f_t * c_{t-1} + i_t * \tilde{c}_t \\ o_t = \sigma\left(W_o \cdot [h_{t-1}, x_t] + b_o\right) \\ h_t = o_t * \tanh\left(c_t\right) \end{cases} \tag{10}$$

where f_t, i_t, and o_t are the activation results of the forget gate, input gate, and output gate, respectively. c_t and h_t are the cell state and the implied state at time t, respectively, W and b are parameter matrices and bias terms. σ and $tanh$ are the activation function of $Sigmoid(\cdot)$ and $Tanh(\cdot)$, whose calculation formula and derivative expression are as follows

$$\begin{cases} \sigma(x) = y = \frac{1}{1+e^{-x}} \\ \sigma'(x) = y(1-y) \end{cases}, \quad \begin{cases} \tanh(x) = y = \frac{e^x - e^{-x}}{e^x + e^{-x}} \\ \tanh'(x) = 1 - y^2 \end{cases}. \tag{11}$$

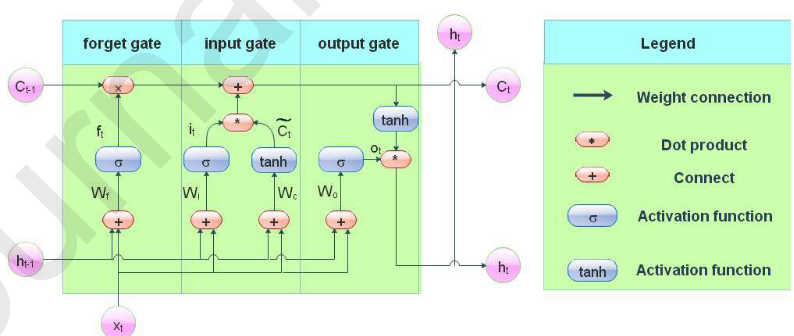


Fig. 1. Architecture of long short-term memory model.

2.3. Improved deep bidirectional long short-term memory neural network

In the framework of RNN, LSTMNN is constructed with LSTM. Its structure is shown in Fig. 2. Replacing the hidden layer in traditional RNN with LSTM to optimize the network structure more precisely. The training process can be expressed as

$$y_t = Vf\left(Ux_t + Wf\left(Ux_{t-1} + Wf\left(Ux_{t-2} + Wf\left(Ux_{t-3} + \ldots\right)\right)\right)\right) \qquad (12)$$

where U, V and W represent the parameter matrix between input layer, LSTM, output layer and h, respectively.

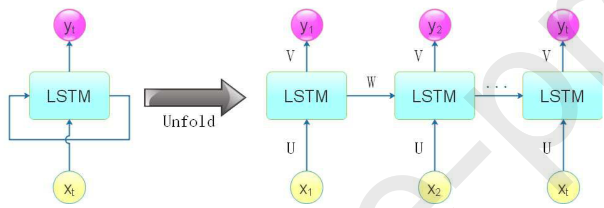


Fig. 2. Topological architecture of LSTM with RNN.

Algorithmic research on deep learning illustrates that increasing the depth of the model can mention the performance of the model. Fig. 3(a) shows a deep LSTMNN that trains the model through three consecutive LSTM blocks. Fig. 3(b) shows a bidirectional independent training neural network model that trains data forward and backward through two separate LSTM blocks to obtain more valid information. An improved deep bidirectional LSTMNN is constructed in combination with deep learning and bidirectional network structure, as shown in Fig. 3(c). The first two LSTM blocks are information correction extraction layer, and both outputs are the input of the third LSTM block. The output after training through the third layer LSTM_3 is passed to the fourth layer LSTM_4 block and the LSTM_2 at the next moment. While increasing the depth of the network, it also improves the capture of effective data information.

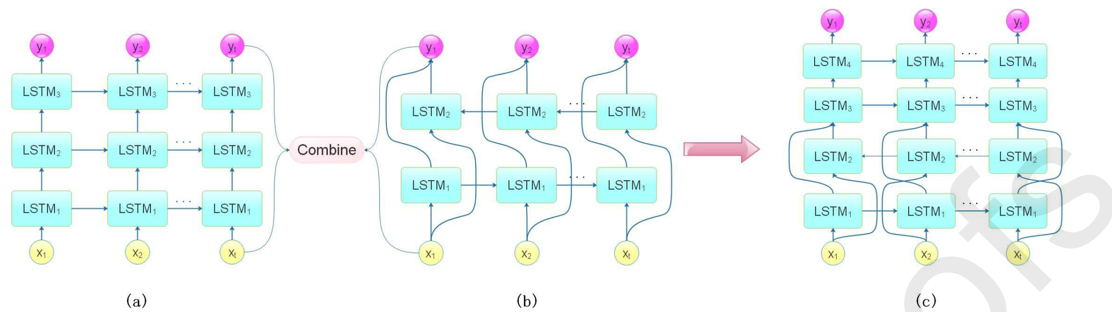


Fig. 3. (a) Architecture of deep LSTMNN. (b) Architecture of bidirectional LSTMNN. (c) Architecture of deep bidirectional LSTMNN.

2.3. Random inheritance formula

Based on geometric Brownian motion, a new random inheritance formula is proposed to train the neural network model (Dufresne, 2001; Gatheral & Schied, 2011). Research reveals that the prediction of financial market price series needs to combine a large number of past data, and historical data has a significant timeliness for future results, that is, recent data has a stronger impact factor than earlier data (Dudek, 2019). Consequently, how to take into account as much as possible the impact factors of historical data on the prediction results, and how to distinguish the impact intensity differences between historical data in different periods reasonably is an important thinking angle to improve the prediction accuracy of the model. Consequently, we propose a random inheritance formula $\psi(t_n)$ based on geometric Brownian motion to solve this problem. Specific definitions are as follows

$$\psi(t_n) = \frac{1}{\zeta} \exp\left\{ \int_{t_c}^{t_n} \mu(t)dt + \int_{t_c}^{t_n} \lambda(t)dB(t) \right\} \tag{13}$$

$$\mu(t) = 1/\exp{(t+a)}, \quad \lambda(t) = \left[\frac{1}{N-1} \sum_{i=1}^{N} (x-\bar{x})^2 \right]^{\frac{1}{2}} \tag{14}$$

where $\zeta(>0)$ represents the time intensity coefficient, t_c is the time when the current data occurs in the training data set, and t_n represents the time of any historical data. $\mu(t)$ is a drift function, which is used to simulate the overall trend of events. Where a in $\mu(t)$ is the forecasting parameter which is equal to the number of samples in the training set. The random inheritance function is used to improve the impact of the latest data, mainly through the drift function. The specific performance is to strengthen the impact weight of recent data and weaken the impact weight of older data. There are many ways to select $\mu(t)$, and the choice of $\mu(t)$ in this paper is the one with the best results after multiple experiments. Combining $\mu(t)$ and Eq. (13), it can be seen that when the difference between t_n and t_c is large, the $\psi(t_n)$ value will decrease, and if the difference between t_n and t_c is small, the $\psi(t_n)$ value will increase. In this way, we can assign appropriate weights based on when the data occurred. $B(t)$ represents standard Brown motion which indicates that the future trajectory of particles in a given time does not depend on the occurrence of past events. $\lambda(t)$ is a volatility function, together with $B(t)$ represents unmeasurable factors. The volatility function is expressed as the variance function of the training data, which reflects the statistical characteristics of the volatility of the price series. Next, the random inheritance formula will be applied to the error correction process of model training.

2.4. Elman recurrent neural network with variational learning rate

Elman neural network is a typical dynamic recurrent neural network (Lasheras et al., 2015; Ruiz, Ruiz, Cuellar, & Pegalajar, 2018). It is based on the structure of back propagation neural network (BPNN) and adds a recurrent layer to the hidden layer. As a consequence, the weight matrix at all times is shared. Fig. 4 indicates a illustrated structure of a multi-input layer ERNN prediction model, in which the number of neuron points in input layer, hidden layer, recursive layer, and output layer is $m * n * n * 1$, the data storage vector of input layer is $x_{it}(i = 1, 2, \cdots, m)$, hidden layer $z_{jt}(i = 1, 2, \cdots, n)$, recurrent layer $u_{jt}(i = 1, 2, \cdots, n)$ at time t, and an output result y_{t+1} at time $t+1$. The forward pass of the ERNN can be defined as follows.

$$\begin{cases} net_{jt} = \sum_{i=1}^{m} w_{ij} x_{it}(k) + \sum_{i=1}^{n} c_{ij} u_{it}(k) \\ u_{jt}(k) = z_{j(t-1)}(k-1) \\ z_{jt}(k) = f_H(\sum_{i=1}^{m} w_{ij} x_{it}(k) + \sum_{i=1}^{n} c_{ij} u_{it}(k)) \\ y_{t+1}(k) = f_T(\sum_{j=1}^{n} v_j z_{jt}(k)) \end{cases} \tag{15}$$

where net_{jt} represents the input of the hidden layer, k represents the number of training times, f_H is the σ activation function, and f_T is an identical mapping.

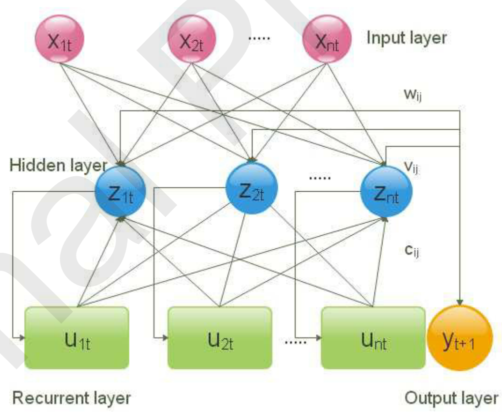


Fig. 4. Topological structure of Elman recurrent neural network.

The choice of learning rate is a crucial parameter in the model training process. A good learning rate will make the model predict more accurate. Experiments have shown that a fixed learning rate is difficult to obtain a better convergence result, and often requires a large number of repeated experiments to select a relatively better one. This increases the uncertainty of the prediction results and is not conducive to the promotion and comparison of the model. Consequently, a variational learning rate arithmetic is proposed, and can be expressed as

$$VR = \alpha \times \Phi(\eta_0, k) \tag{16}$$

where α is the volatility constant, η_0 is the initial learning rate, and $\Phi(\eta_0, k)$ can be expressed as

$$\Phi(\eta_0, k) = \eta_0 * \exp\{-\log_{10}(k)\} + \frac{1}{\gamma} * normrnd(0,1) \tag{17}$$

where $normrnd(0, 1)$ represents a random sample point obeying the standard normal distribution with mean value of 0 and standard deviation of 1, which can greatly increase the learning rate at a certain time, thus avoiding the case where the conventional exponential decay learning rate falls into a local optimal solution because the learning rate is too small. Through the variational learning rate arithmetic, it is possible to flexibly obtain a more reliable and appropriate learning rate and improve the prediction accuracy of the model.

2.5. Error correction algorithm based on random inheritance formula

In this section, we apply the RIF to the error correction process of model training, optimize the parameter update process, and improve the prediction accuracy. Take the LSTM model as an example, the specific process is as follows.

Eq. (10) gives the forward propagation process of LSTM, in which W_f, W_i, W_c and W_o are merged and can be split into

$$\begin{cases} W_f = W_{fx} + W_{fh} \\ W_i = W_{ix} + W_{ih} \\ W_{\overline{c}} = W_{cx} + W_{ch} \\ W_o = W_{ox} + W_{oh} \end{cases} \tag{18}$$

where W_{ij} denotes that W_{ij} is the connection matrix from j to i. The input and output of the forget gate, input gates and output gate can be expressed as

$$o_t = \sigma\left(net_{o,t}\right), \quad net_{o,t} = W_{oh}h_{t-1} + W_{ox}x_t + b_o \tag{19}$$

$$f_t = \sigma\left(net_{f,t}\right), \quad net_{f,t} = W_{fh}h_{t-1} + W_{fx}x_t + b_f \tag{20}$$

$$i_t = \sigma\left(net_{i,t}\right), \quad net_{i,t} = W_{ih}h_{t-1} + W_{ix}x_t + b_i \tag{21}$$

$$\tilde{c}_t = \tanh\left(net_{\tilde{c},t}\right), \quad net_{\tilde{c},t} = W_{ch}h_{t-1} + W_{cx}x_t + b_c. \tag{22}$$

The error of the n_{th} sample point at time t_n is defined as

$$E_{t_n} = \frac{1}{2}\varepsilon_{t_n}^2 = \frac{1}{2}\left(o_{t_n} - y_{t_n}\right)^2. \tag{23}$$

Combining RIF with error correction process, a new definition of error E_{t_n} can be obtained, which can be expressed as follows.

$$E_{t_n} = \frac{1}{2}\psi(t_n)\varepsilon_{t_n}^2 = \frac{1}{2}\psi(t_n)(o_{t_n} - y_{t_n})^2 \tag{24}$$

where $\psi(t_n)$ denotes the RIF, which will assign corresponding weight on the basis of the time of sample occurrence. Then the global error can be defined as

$$E = \frac{1}{N}\sum_{n=1}^{N} E_{t_n} = \frac{1}{2N}\sum_{n=1}^{N}\frac{1}{\gamma}\exp\left\{\int_{t_c}^{t_n}\mu(t)dt + \int_{t_c}^{t_n}\sigma(t)dB(t)\right\}(s_{t_n} - y_{t_n})^2. \tag{25}$$

Applying the newly defined error function E, the parameter matrix is updated by gradient descent method (Abiyev, 2011). First, the partial derivatives of the error function with respect to each parameter are obtained, and then the parameters are updated. The specific process is as follows.

$$\frac{\partial E}{\partial W_{fx}} = \frac{\partial E}{\partial net_{f,t}}\frac{\partial net_{f,t}}{\partial W_{fx}} = \delta_{f,t}\psi(t)x_t, \quad \frac{\partial E}{\partial W_{fh}} = \sum_{j=t_0}^{t_n}\delta_{f,j}\psi(t)h_{j-1} \tag{26}$$

$$\frac{\partial E}{\partial W_{ix}} = \frac{\partial E}{\partial net_{i,t}} \frac{\partial net_{i,t}}{\partial W_{ix}} = \delta_{i,t}\psi(t)x_t, \quad \frac{\partial E}{\partial W_{ih}} = \sum_{j=t_0}^{t_n} \delta_{i,j}\psi(t)h_{j-1} \tag{27}$$

$$\frac{\partial E}{\partial W_{ix}} = \frac{\partial E}{\partial net_{i,t}} \frac{\partial net_{i,t}}{\partial W_{ix}} = \delta_{i,t}\psi(t)x_t, \quad \frac{\partial E}{\partial W_{ch}} = \sum_{j=t_0}^{t_n} \delta_{\tilde{c},j}\psi(t)h_{j-1} \tag{28}$$

$$\frac{\partial E}{\partial W_{ox}} = \frac{\partial E}{\partial net_{o,t}} \frac{\partial net_{o,t}}{\partial W_{ox}} = \delta_{o,t}\psi(t)x_t, \quad \frac{\partial E}{\partial W_{oh}} = \sum_{j=t_0}^{t_n} \delta_{o,j}\psi(t)h_{j-1} \tag{29}$$

where δ represents the error term, the specific definition can be known in (Hochreiter & Schmidhuber, 1997). Next, the parameters are corrected by the gradient descent method, and the updating rule of the parameter matrix W_{ij} is

$$W_{ij} = W_{ij} - \eta \cdot \frac{\partial E}{\partial W_{ij}} \tag{30}$$

where η is learning rate. With continuous iteration, the parameter matrix is updated continuously, so that the output results are closer to the real data.

2.6. Integrated framework of the proposed model

A novel hybrid model is proposed, which combines EWT feature extraction algorithm, DBLSTMNN, ERNN, RIF-based error correction algorithm and variational learning rate. Its concrete construction and application process are shown in Fig. 5. We realize the feature sequences extraction of the original time series by EWT through Matlab R2018b. The decomposition advantage of EWT makes it excellent in the extraction of feature sequences. Different models are trained and predicted according to the frequency of feature sequences. For the low-frequency sequence, we choose the RIF-DBLSTMNN model, and for high-frequency sequences, we choose the RIF-VRERNN model. Both models are introduced in Sections 2.2-2.5. Considering the timeliness of historical data in the energy futures market, the RIF error correction method is proposed and applied to model training in order to obtain better training parameters. The RIF-DBLSTMNN model is built by Tensorflow. The number of moving windows is defined as 5, and the cross-validation method is used in the model training process. Hyper parameters of the model are selected by calculating the root-mean-square error (RMSE) between the prediction results and real data and visualizing the data results. In the paper, the batch size is 32 and the hidden size is 30. The training and prediction of RIF-VRERNN are realized through Matlab. Considering the sensitivity of the ERNN model to the learning rate, the variational learning rate is proposed to adjust the value of the learning rate according to the training process. The parameter γ is 1000. In this paper, the number of neuron nodes in the input layer, hidden layer and recurrent layer of the ERNN model are set to 4, 9, and 9. Finally, the prediction results of the original time series are obtained after accumulating all the prediction results of the feature sequences. The detailed steps for applying the proposed EWT-RIF-DBLSTM-VRERNN model are shown below.

(i) Feature extraction. The EWT algorithm is used to extract the feature sequences of original time series $X(t)$. In this paper, the original time series is decomposed to obtain six intrinsic mode functions (IMF_s).

(ii) Model building. Combined with the introduction of methods in Sections $2.2-2.5$, the RIF-DBLSTMNN and RIF-VRERNN models are constructed. The training and prediction of RIF-DBLSTMNN and RIF-VRERNN models are implemented by Matlab R2018b and Tensorflow respectively.

(iii) Model prediction. First, apply the RIF-DBLSTMNN model to predict the low-frequency feature sequence IMF_1, and get the prediction result $\widehat{\mathrm{IMF}_1}$. Then use the RIF-VRERNN model to predict the remaining high-frequency feature sequences IMF_i ($2 \leq i \leq 6$), and the prediction results $\widehat{\mathrm{IMF}_i}$ ($2 \leq i \leq 6$) are obtained.

(iv) Aggregate the predicted results of all feature sequences to obtain an accumulated value $\widehat{X(t)}$, which represents the predicted value of original time series. The specific formula can be expressed as

$$\widehat{X(t)} = \sum_{i=1}^{6} \widehat{\mathrm{IMF}}_i. \tag{31}$$

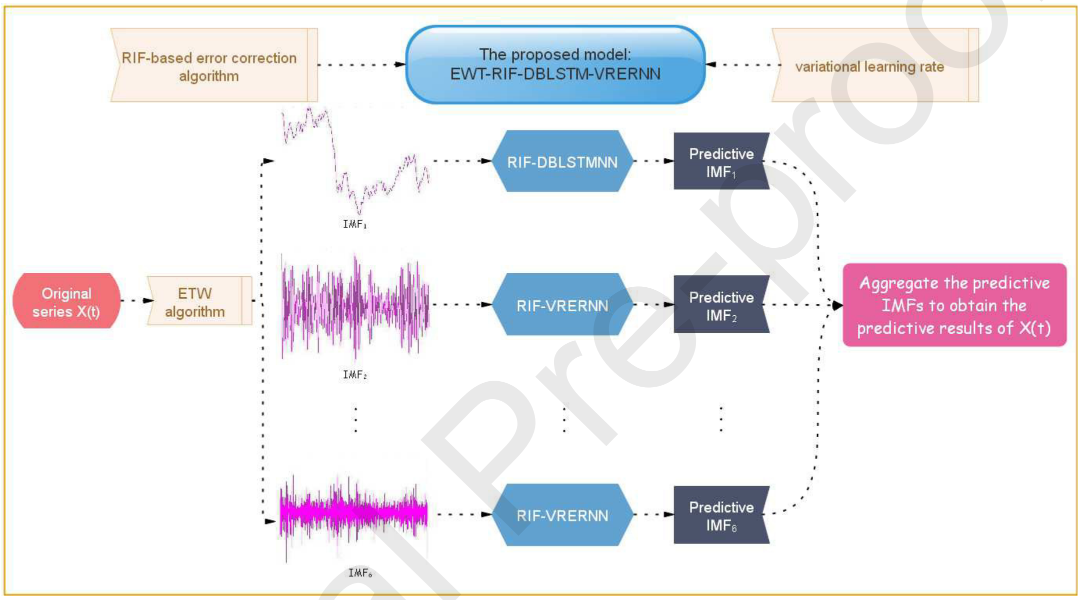


Fig. 5. Integrated framework of multi-hybrid model.

3. Performance evaluation indexes

3.1. Basic accuracy metrics

In order to further study the predictive abilities of the proposed model RIF-DBLSTMNN, some regression model evaluation indicators are utilized in this work. Let N denotes the total number of samples, d_t denotes the true value, y_t denotes the predicted value and $\overline{d}$ denotes the mean of samples. The selection of indicators and their formulas are listed in Table 1 (Hsu, 2013; Khandelwal, Adhikari, & Verma, 2015; Mo, Wang, & Niu, 2016; Tumer, Kocer, & Koca, 2017; Zhang et al., 2018).

Table 1. Selection and introduction of multiple evaluation indicators.

Comprehensive evaluation indicators	
Mean absolute error (MAE)	$\frac{1}{N}\sum_{t=1}^{N}\lvert d_t - y_t\rvert$
Root mean square error (RMSE)	$\sqrt{\frac{1}{N}\sum_{t=1}^{N}(d_t - y_t)^2}$
Symmetric mean absolute percentage error (SMAPE)	$100 \times \frac{2}{N}\sum_{t=1}^{N}\frac{\lvert d_t - y_t\rvert}{\lvert d_t\rvert + \lvert y_t\rvert}$
Theil inequality coefficient (TIC)	$\frac{\sqrt{\frac{1}{N}\sum_{t=1}^{N}(d_t - y_t)^2}}{\sqrt{\frac{1}{N}\sum_{t=1}^{N}{d_t}^2}+\sqrt{\frac{1}{N}\sum_{t=1}^{N}{y_t}^2}}$
R-squared (R^2)	$1 - \frac{\sum_{t=1}^{N}(d_t - y_t)^2}{\sum_{t=1}^{N}(d_t - \overline{d})^2}$

3.2. Complex accuracy metrics

In this section, an efficient evaluation algorithm q-order dyadic scales complexity invariant distance (q-DSCID) is presented for energy futures price time series, which can test and quantify the correlation between different time series. Therefore, we can apply it to weigh the predictions. Suppose two time series $D = \{d_1, d_2 \cdots, d_n\}$, $Y = \{y_1, y_2 \cdots, y_n\}$. The distance between two sets of sequences is defined as follows

$$\mathrm{ED}^q(D, Y) = \left[\sum_{i=1}^{n}(d_i - y_i)^q\right]^{\frac{1}{q}}. \tag{32}$$

The definition of invariant distance can be obtained from the following correlation factors and the distance $\mathrm{ED}^q(D, Y)$ between two sets of time series D and Y

$$q - \mathrm{CID}(D, Y) = \mathrm{ED}^q(D, Y) \times \mathrm{CF}^q(D, Y) \tag{33}$$

$\mathrm{CF}^q(D, Y)$ is a complexity correlation factor, which is expressed as

$$\mathrm{CF}^q(D, Y) = \frac{\max\{\mathrm{CE}^q(D), \mathrm{CE}^q(Y)\}}{\min\{\mathrm{CE}^q(D), \mathrm{CE}^q(Y)\}}. \tag{34}$$

The complexity estimates of time series $T = \{t_1, t_2 \cdots, t_n\}$ are defined as

$$\mathrm{CE}^q(T) = \left[\sum_{i=1}^{n-1}(t_{i+1} - t_i)^q\right]^{\frac{1}{q}}. \tag{35}$$

The q-DSCID evaluation algorithm combines multiple time scales and multiple displacement scale while evaluating the predictions, which can be expressed as

$$q - \mathrm{DSCID} = (D, Y, \tau, S) = \frac{1}{S}\sum_{s=1}^{S} q - \mathrm{CID}(D(\tau, s), Y(\tau, s)). \tag{36}$$

Dyadic scales time series are given by

$$T(\tau, s) = \{t_1(\tau, s), t_2(\tau, s), \cdots, t_n(\tau, s)\} \tag{37}$$

where the element is represented as

$$t_j(\tau, s) = \frac{1}{\tau}\sum_{i=(j-1)\tau+s}^{j\tau+s-1} x_i, \quad j = 1, 2, \cdots, (n - S)/\tau. \tag{38}$$

Different from the previous regression evaluation indexes, the q-CID distance has a close relationship with the correlation complexity of time series D and Y. The q-CID will degenerate into the traditional CID distance when $q = 2$, and the $\mathrm{ED}^q(D, Y)$ distance becomes the universal Euclidean distance. The q-CID distance can be scaled by determining the value of q to distinguish the difference between time series more clearly, especially when the difference between time series is subtle. q-DSCID considers time scale parameter τ and displacement scale parameter S, so we can utilize q-DSCID to test the discrepancy between D and Y in different time scales and displacement scales. According to the equation of $t_j(\tau, s)$, the action process of scale τ and S can be regarded as the coarse graining process of the initial sequence, and the scale τ determines the degree of coarse graining. Without considering the displacement scale S, the initial sequence is obtained especially when $\tau = 1$. After combining the displacement scale S, the starting position of the coarse graining process can be set, and q-DSCID can be regarded as the average value of q-CID($D(\tau$,$s)$,$Y(\tau$,$s)$) under different displacement scales. Thus, we can see that q-DSCID is a more comprehensive and reliable error evaluation algorithm.

4. Numerical experiment on empirical analysis and comparison

4.1. Data selection and processing

A total of two energy market futures price data sets are selected: the crude oil futures data of West Texas Intermediate crude oil (WTI), ICE Brent crude oil (BRE). Many investors regard the WTI spot contracts as the benchmark price for measuring crude oil price changes in the international energy market. Brent crude oil futures is the most widely applied contract in global crude oil pricing. At present, about seventy percent of spot crude oil trading pricing refers to Brent scale. Some Asian crude oil producers have recently shifted from their regional scale to Brent, and there are two-thirds of the world's oil market crude oil prices related to Brent crude oil, including China's refined oil price reference mechanism. Therefore, it is also meaningful to study their futures prices. The daily closing prices of WTI and BRE in consecutive trading days over a period of time are selected as the sample data set. All data sets can be obtained from the website (https://cn.investing.com/). The selection and division of the data set is shown in Table 2.

Table 2. Data set selection and division.

Data set	Training	Test	Start date	Expiry Date
WTI	1500	500	2012.01.19	2019.09.14
BRE	1500	500	2011.12.19	2019.09.18

Statistical characteristics of the logarithmic returns series are explored for the selected series. Define $F(t)$ to represent the crude oil futures price of the selected data set at time t, then its logarithmic return $r(t)$ can be expressed as

$$r(t) = \ln F(t+1) - \ln F(t). \tag{39}$$

The statistical characteristics of returns series are calculated: mean, median (Med), variance (Var), standard deviation (SD), maximum (Max), minimum (Min), skewness (Skew) and kurtosis (Kurt) which are shown in Table 3. In addition, the autocorrelation and conditional heteroscedasticity tests are performed on the returns series. Select the residual squared correlation test method to realize the conditional heteroscedasticity test of the returns series. Table 4 shows the autocorrelation coefficient (ACF) values with a lag order of four. Where ACF_i represents the value of the autocorrelation coefficient when the lag order is i. The values of the autocorrelation coefficients obtained by the autocorrelation test are very close to 0 and fluctuate around the value of 0, indicating that there is no obvious autocorrelation in the returns series. Similarly, the values of the autocorrelation function obtained by the residual squared correlation test are mostly close to 0, indicating the existence of heteroscedasticity, that is, there is a significant ARCH effect in the returns series.

Table 3. Statistical characteristics of returns series.

		Statistical characteristics							
		Mean	Med	Var	SD	Max	Min	Skew	Kurt
WTI	All samples	-2.3388e-04	6.0054e-04	4.2580e-04	0.0206	0.1369	-0.0907	0.2160	6.6273
	Training	-4.3624e-04	1.4996e-04	4.3731e-04	0.0209	0.1162	-0.0907	0.2823	5.9845
	Testing	3.7444e-04	0.0021	3.9154e-04	0.0198	0.1369	-0.0823	-0.0089	9.0273
BRE	All samples	-2.4443e-05	3.7380e-04	3.8126e-04	0.0195	0.1364	-0.0886	0.2200	6.9751
	Training	-3.9928e-04	-3.8143e-04	3.8442e-04	0.0196	0.1042	-0.0886	0.2889	6.1613
	Testing	2.7104e-04	0.0021	3.7222e-04	0.0193	0.1364	-0.0744	0.0050	9.6206

Table 4. Autocorrelation and conditional heteroscedasticity tests for returns series.

Data set	Autocorrelation test				Conditional heteroscedasticity test			
	ACF_1	ACF_2	ACF_3	ACF_4	ACF_1	ACF_2	ACF_3	ACF_4
WTI	-0.0724	0.0111	9.7059e-04	0.0297	0.2275	0.1988	0.1882	0.1312
BRE	-0.0832	0.0241	0.0043	0.0083	0.2136	0.1310	0.2208	0.1099

To eliminate the adverse effects caused by singular sample data, we filter and normalize the data set appropriately. The specific steps are as follows. (i) The 3-σ principle is applied to eliminate the data whose mean fluctuates above or below three times the standard deviation. Taking into account the impact of random interference on crude oil price fluctuations, this method can effectively eliminate random interference. This not only retains the development trend and status of the original data, but also improves the prediction accuracy of the model. (ii) In the process of model training and prediction, we normalize the input series with the method of physical deviation standardization whose conversion function is as follows

$$\tilde{I}(t) = \frac{I(t) - \min I(t)}{\max I(t) - \min I(t)} \tag{40}$$

where $I(t)$ stands for the input series, $\min I(t)$ and $\max I(t)$ represent the lowest value and the highest value of training set separately. After we get the output value, the real data can be retrieved through

$$I(t) = \tilde{I}(t)(\max I(t) - \min I(t)) + \min I(t). \tag{41}$$

4.2. Implement by the proposed model

The feature sequence IMF_i $(i = 1, 2, 3, 4, 5, 6)$ is extracted by applying the EWT algorithm to the initial price series, where IMF_1 is a low frequency feature sequence, which reflects the overall fluctuation trend of the data, and the rest are high frequency feature sequences that reflect the local detail features of the price series. Fig. 6 shows the sequences of features after BRE and WTI are extracted, respectively. First, using the RIF-DBLSTMNN to predict the low frequency sequence IMF_1, the predicted result is shown in Fig. 7(a). The prediction results of the high frequency sequences predicted by RIF-VRERNN are shown in Figs. 7(b)(c)(d)(e)(f). In the legend, the “Predictive data” refers to the result $\widehat{\mathrm{IMF}}_i$ obtained by applying model training and predicting IMF_i, while the “Actual date” refers to the feature sequence IMF_i obtained after EWT feature extraction. The prediction results of the feature sequences are accumulated to obtain the prediction result of the initial price series.

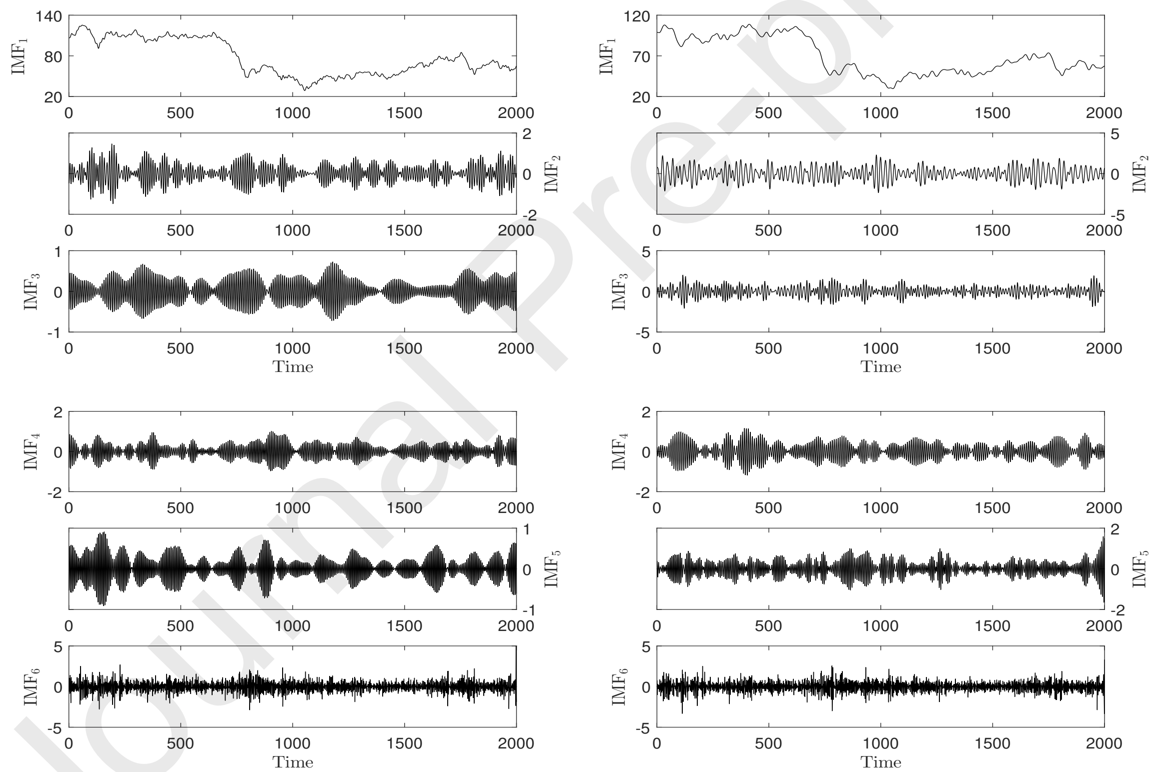
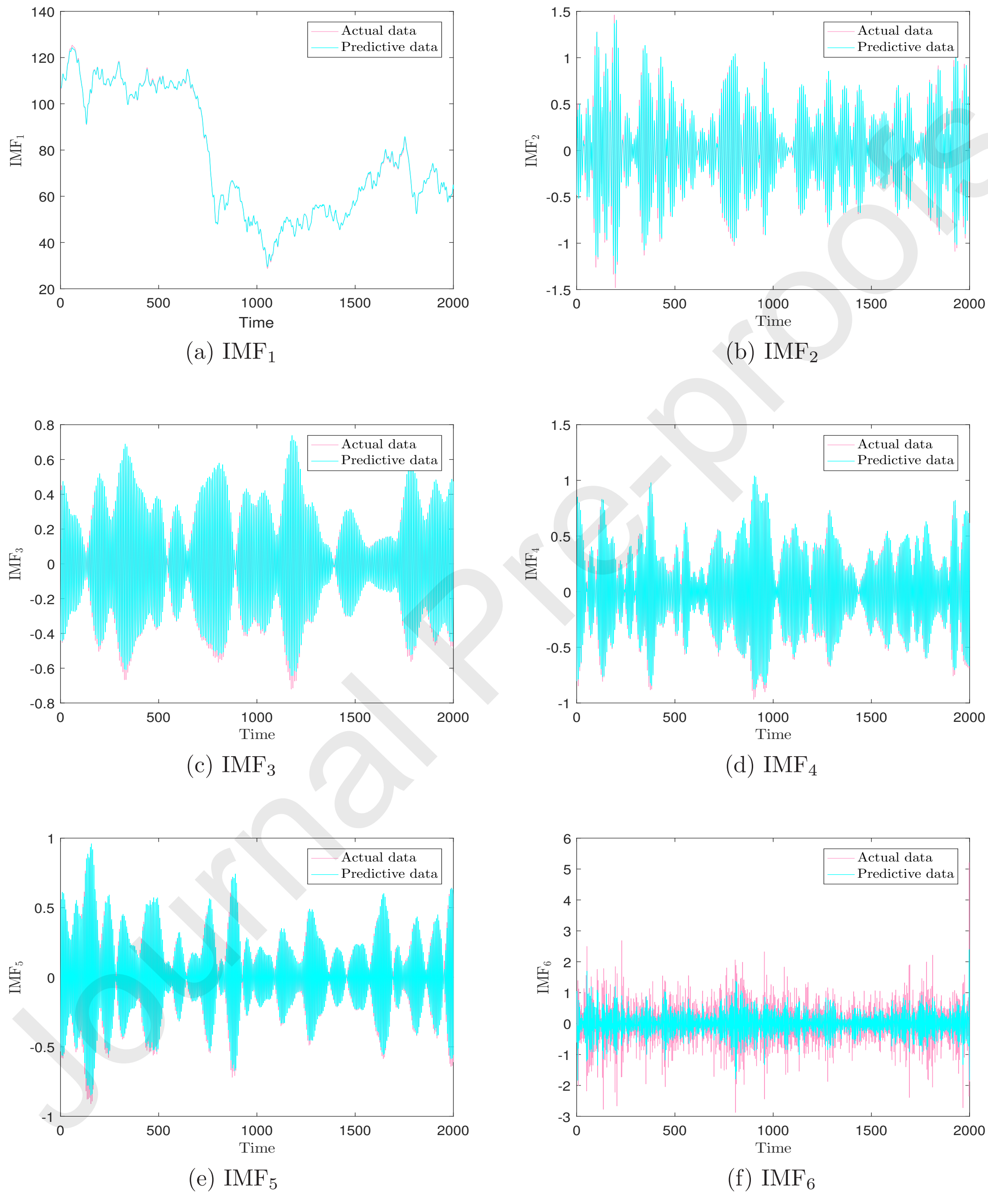


Fig. 6. (a) Empirical wavelet transform decomposition results of BRE. (b) Empirical wavelet transform decomposition results of WTI.



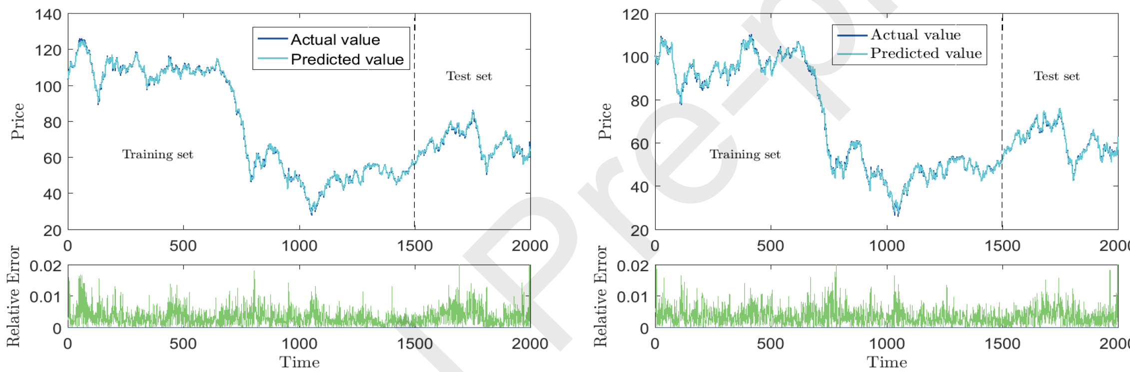
(a) IMF_1 (b) IMF_2

(c) IMF_3 (d) IMF_4

(e) IMF_5 (f) IMF_6

Fig. 7. (a) Predictions of low frequency sequence of BRE by RIF-DBLTMNN. (b)(c)(d)(e)(f) Predictions of high frequency sequences of BRE by RIF-VRERNN.

Fig. 8 shows the predicted results of the proposed multi-hybrid EWT-RIF-DBLSTM-VRERNN model for BRE and WTI. Simultaneously, the absolute correlation error map of the prediction results is also shown in Fig. 8, which can be expressed as $RE(t) = |d_t - y_t|/d_t$ (Liao & Wang, 2010). It can be seen that the predicted results are basically coincident with the fluctuations of the actual data. The values of RE are also concentrated in $(0, 0.01)$, and only a few local data points exceed 0.01 but less than 0.02. The prediction accuracy of the model from the image is excellent.

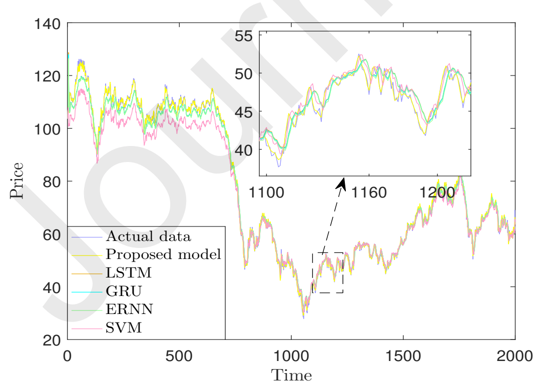


(a) BRE (b) WTI

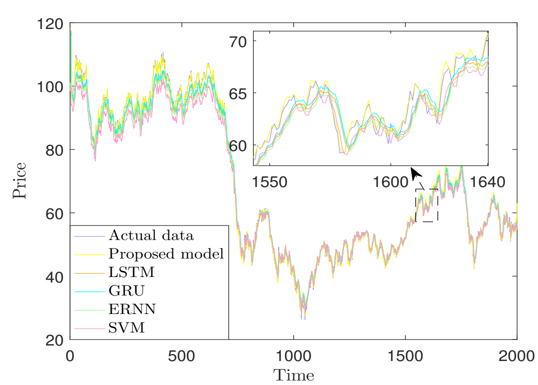
Fig. 8. The predicted results of the proposed model for the initial sequence.

4.3. Forecast by different comparison models

Four sets of individual models GRU, LSTM, ERNN, and SVM are selected as comparison models and compared with the proposed hybrid model (Wu, Liao, Miao, & Du, 2019; Chen, Jing, Chang, & Liu, 2019). The prediction results of WTI and BRE obtained by applying each model are shown in Fig. 9. It can be seen that the prediction of the SVM is significantly different from the data, and the prediction result of the hybrid model is better than any comparison models. Linear regression analysis is performed on the prediction results of each model. Linear regression is a statistical analysis method used to determine the quantitative relationship between two or more variables. We use linear regression to explore the dependence between model predictions and real data. Set the actual data as x-axis and the predicted result as y-axis, draw a scatterplot, and calculate the best linear fit equation. The closer the slope of the fitted equation is to 1, indicating that the deviation between the actual value and the predicted value is not significant and the model prediction result is more accurate. The regression analyses of each model are shown in Fig. 10 and Fig. 11. It is worth noting that the data points of the proposed model are basically distributed on the fitting function, indicating that the prediction results are very close to the real data. The data points of other models are relatively scattered, and there is a certain distance from the fitting function. Table 5 gives the parameter values for the regression equation $y = ax + b$ for each model. The regression equation parameters of the proposed model for WTI are $a = 1.0278$, $b = 1.6410$, which is closest to the ideal case $y = x$, followed by the LSTM model $a = 0.9653$, $b = 2.1486$, and the SVM is the worst with $a = 0.9078$, $b = 5.0779$.

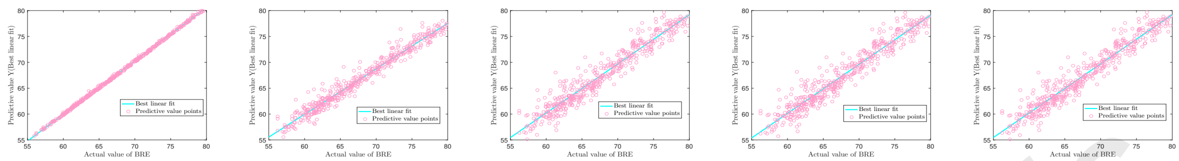


(c) BRE



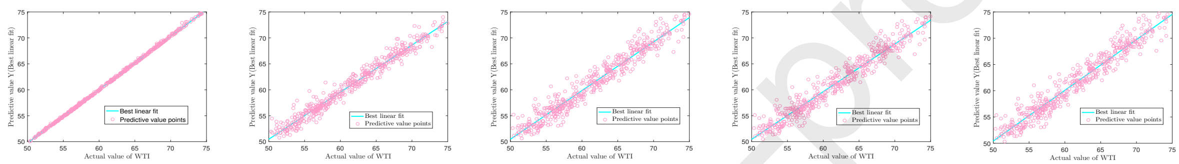
(d) WTI

Fig. 9. Comparison of predicted results between different models and the proposed model.



(a) (b) (c) (d) (e)

Fig. 10. Linear regression analysis of BRE prediction results for each model.



(a) (b) (c) (d) (e)

Fig. 11. Linear regression analysis of WTI prediction results for each model.

Table 5. Linear regression parameters for different models.

Model	WTI		BRE	
	a	b	a	b
SVM	0.9078	5.0779	0.8802	7.1251
GRU	0.9379	3.5764	0.9498	2.5402
ERNN	0.9182	4.5433	0.9493	3.0770
LSTM	0.9653	2.1486	0.9643	2.4728
The proposed model	1.0278	-1.6410	1.0307	-1.6820

4.4. Comparison and analysis for models

In this section we will compare the proposed EWT-RIF-DBLSTM-VRERNN model from multiple angles, analyze the superiority of the model and the applicability of the hybrid arithmetic. From the prediction results of comparison models, we calculated their underlying error metrics and the q-DSCID values at $\tau = 1$. Table 6 illustrates the error metrics for each model's BRE predictions. The R^2 value of the proposed model is the largest approximation equal to one, and the values of MAE, TIC, RMSE, SMAPE, and q-DSCID with $\tau = 1$ are 0.2454, 0.0024, 0.3325, 0.3419, and 7.6375, respectively, which are the smallest and are significantly better than the prediction results of other models. There is no doubt that the proposed hybrid model has more outstanding performance than other models. Table 7 reveals the error measurement results of the WTI prediction results for each model. The hybrid model still has a large lead, while the results of LSTM and GRU are not much different between the two sets of data, and the SVM indicators are the worst.

Table 6. Evaluation indicators of distinct forecasting models for BRE.

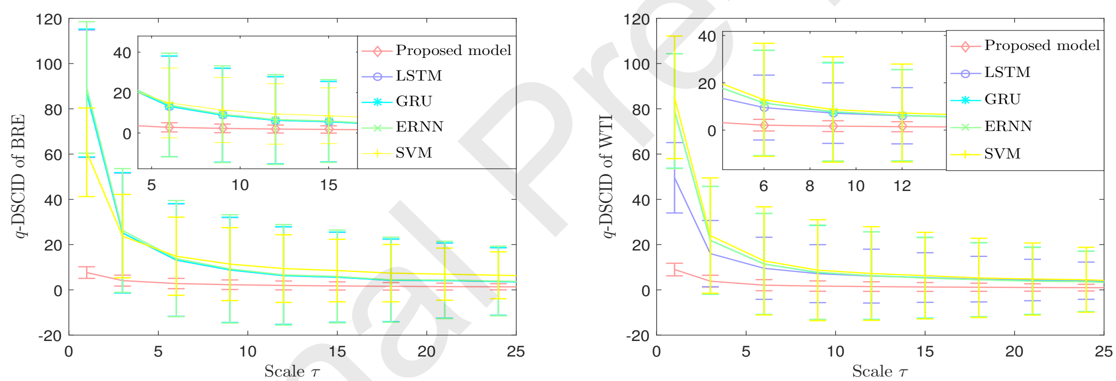
Model	Metrics					
	R^2	MAE	TIC	RMSE	SMAPE	q-DSCID
Proposed model	0.9977	0.2454	0.0024	0.3325	0.3419	7.6375
LSTM	0.9369	1.3185	0.0126	1.7261	1.9672	86.6826
ERNN	0.9365	1.4321	0.0129	1.7512	2.0837	88.8088
GRU	0.9366	1.3242	0.0127	1.7312	1.9754	86.9891
SVM	0.9318	1.3870	0.0132	1.7965	2.0703	89.4513

Table 7. Evaluation indicators of distinct forecasting models for WTI.

Model	Metrics					
	R^2	MAE	TIC	RMSE	SMAPE	q-DSCID
Proposed model	0.9966	0.2085	0.0032	0.3933	0.3329	8.9861
LSTM	0.9580	1.0831	0.0114	1.3942	1.7729	49.5346
ERNN	0.9413	1.2603	0.0134	1.6425	2.1009	78.9575
GRU	0.9445	1.2410	0.0130	1.5979	2.0648	79.0283
SVM	0.9384	1.3335	0.0138	1.6865	2.2117	85.0840

Further, the complex accuracy metric q-DSCID is applied to evaluate the performance of the proposed model. Choose S=mean$\{\tau\}$, τ varies from $1, 2, 3, \cdots, 40$, and every model are predicted 50 times respectively. The q-DSCID values of each model at different time scales are calculated, and draw the q-DSCID results of different models at different time scales τ with error bar in Fig. 12. It is shown that when the initial scale $\tau = 1$, the

q-DSCID value of the proposed model is the smallest in all the models, which is obviously superior to other models. With the increase of time scale τ, the q-DSCID value gap between the models gradually decreases and tends to stabilize. As the time scale increases, the q-DSCID value of the proposed model is always the smallest. The q-DSCID value of the SVM is always the largest. The q-DSCID values of the ERNN, GRU, and LSTM are not much different, and basically coincide, but the values of the LSTM and GRU are still slightly smaller than the ERNN. It can be considered that the proposed hybrid neural network model is the best and then LSTM, GRU, ERNN, SVM.



(a) BRE (b) WTI

Fig. 12. Variation of q-DSCID values of each model on scale τ.

Table 8 shows the error metrics of WTI and BRE prediction results using RIF-DBLSTMNN and RIF-VRERNN separately after EWT decomposition. The prediction results using one model alone are significantly less accurate than the multiple hybrid prediction models. The prediction results obtained using the RIF-DBLSTM predictive feature sequence alone are superior to the RIF-VRERNN alone. The reason is that the LSTM model is more suitable for low-frequency sequence prediction and the ERNN model is more suitable for high-frequency prediction, while the low-frequency sequence contains the overall fluctuation trend of the initial sequence. By classifying the feature sequences and exploiting more suitable models to predict, the prediction accuracy of the model can be ameliorated.

Table 8. Evaluation indicators of multi-model vs. single model prediction for BRE and WTI.

	Metrics					
	R^2	MAE	TIC	RMSE	SMAPE	q-DSCID
BRE						
Proposed model	0.9977	0.2454	0.0024	0.3325	0.3419	7.6375
EWT-RIF-DBLSTMNN	0.9849	0.6287	0.0062	0.8451	0.9206	37.5283
EWT-RIF-VRERNN	0.9534	1.1559	0.0108	1.4836	1.7218	51.3974
WTI						
Proposed model	0.9966	0.2085	0.0032	0.3933	0.3329	8.9861
EWT-RIF-DBLSTMNN	0.9899	0.5286	0.0055	0.6824	0.8722	27.3806
EWT-RIF-VRERNN	0.9747	0.8520	0.0088	1.0787	1.4160	32.6937

Tables 9 displays the error metric results of the prediction results of each model after combining the RIF-based error correction algorithm. Combined with Tables 6 and 7, it is clear that the prediction accuracy of the improved model has been greatly improved. After LSTM is combined with the error correction algorithm, the metrics are greatly improved, and its R^2, MAE, TIC, RMSE, SMAPE, and q-DSCID indicators increased by 5.657%, 57.679%, 40.317%, 59.956%, 58.728%, and 78.397%, respectively. However, there is still a gap between the prediction accuracy of each model and the proposed model. It is worth mentioning that although the prediction accuracy of RIF-DBLSTMNN is still not as good as the proposed model, it is very close to the real data.

Table 9. Metrics of BRE and WTI predictions with RIF error correction algorithm.

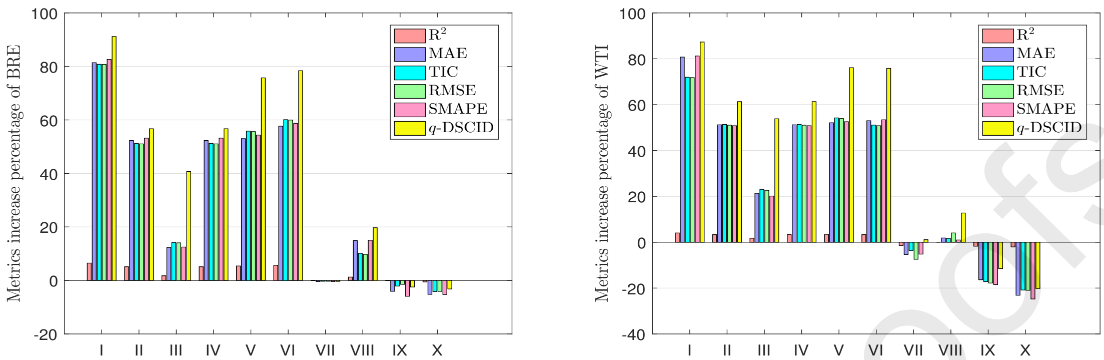
	Metrics					
	R^2	MAE	TIC	RMSE	SMAPE	q-DSCID
BRE						
Proposed model	0.9977	0.2454	0.0024	0.3325	0.3419	7.6375
RIF-DBLSTMNN	0.9916	0.4872	0.0046	0.6316	0.7061	20.4748
RIF-LSTMNN	0.9899	0.5580	0.0050	0.6911	0.8119	18.7261
RIF-ERNN	0.9487	1.1217	0.0114	1.5577	1.6717	69.5989
WTI						
Proposed model	0.9966	0.2085	0.0032	0.3933	0.3329	8.9861
RIF-DBLSTMNN	0.9941	0.4010	0.0042	0.5211	0.6517	15.3421
RIF-LSTMNN	0.9898	0.5092	0.0056	0.6858	0.8264	17.1263
RIF-ERNN	0.9551	1.0714	0.0117	1.4380	1.7857	61.7940

In order to more clearly compare the superiority of the EWT-RIF-DBLSTM-VRERNN, combined with Tables 6-9, the model accuracy metric increase percentage graph based on LSTM accuracy is illustrated in Fig. 13. The way increase percentage (IP) is constructed can be expressed as

$$IP = \begin{cases} \frac{Metrics_{Others} - Metrics_{LSTM}}{Metrics_{LSTM}} & Others = R^2 \\ -\frac{Metrics_{Others} - Metrics_{LSTM}}{Metrics_{LSTM}} & Otherwise \end{cases} \tag{42}$$

where $Metrics_{LSTM}$ represents the accuracy metrics of the LSTM, and $Metrics_{Others}$ represents the accuracy metrics of other models.

Fig. 13 shows the IP map of each model based on LSTM metrics for BRE and WTI. The numbers I, II, $\cdots$, X represent the models EWT-RIF-DBLSTM-VRERNN, EWT-RIF-DBLSTMNN, EWT-RIF-VRERNN, DBLSTMNN, RIF-LSTMNN, RIF-DBLSTMNN, GRU, RIF-ERNN, ERNN, SVM. We can draw the following conclusions. (i) The novel multi-hybrid neural network model has the best prediction accuracy among many model models, and the IP values of each indicator are the largest. (ii) Deep bidirectional training structure and RIF-based error correction algorithm can effectively improve the prediction accuracy of the model. (iii) After extracting the feature sequence by EWT algorithm, the performance of the model can be effectively improved by using appropriate prediction models.



(a) BRE (b) WTI

Fig. 13. Percent histogram of metrics improvement based on LSTM accuracy.

On the basis of one-step ahead prediction, considering that multi-step ahead forecasting has extremely important value in many fields, we make 2-step and 3-step ahead prediction of the proposed model. Multi-step ahead forecasting can obtain forecast values at multiple consecutive time points in the future, and is suitable for forecasting the mid-to-long term situation. We choose the direct multi-step prediction strategy to achieve 2-step and 3-step ahead forecasting (Bontempi, Taieb, & Borgne, 2013). The evaluation metrics for the prediction results are shown in Table 10. It can be seen that the deviation between the multi-step ahead forecasting results and the real data is not significant. Obviously, the result of the 2-step ahead forecasting is worse than that of one-step ahead forecasting, and better than that of 3-step ahead forecasting. It can be concluded that in most cases, for the same prediction model, as the number of prediction steps increases, its prediction performance will show a downward trend.

Table 10. Metrics of multi-step ahead forecasting for BRE and WTI.

	Metrics					
	R^2	MAE	TIC	RMSE	SMAPE	q-DSCID
BRE						
1-step ahead	0.9977	0.2454	0.0024	0.3325	0.3419	7.6375
2-step ahead	0.9852	0.4811	0.0054	0.6012	0.7287	19.6418
3-step ahead	0.9585	0.8673	0.0094	1.1341	1.3418	47.7261
WTI						
1-step ahead	0.9966	0.2085	0.0032	0.3933	0.3329	8.9861
2-step ahead	0.9887	0.4912	0.0053	0.6425	0.7168	22.5538
3-step ahead	0.9482	1.0367	0.0125	1.2538	1.6375	54.8193

In order to judge the predictive ability of different models more rigorously (Diebold & Mariano, 1995), superior predictive ability (SPA) test method proposed by Hansen (2005) based on out of sample rolling window is applied in this paper to test the prediction ability of the proposed model. Generally used ERNN, GRU, LSTM models and the proposed model are selected for SPA test. Two most commonly used loss functions MSE and MAE are applied as the evaluation criteria for prediction accuracy to calculate the p value of the SPA test. Due to space limitations, more detailed content can be obtained in Hansen (2005). Under a certain loss function L, if the p value of the SPA test of the basic model is larger than that of other models, it indicates that the prediction accuracy of the basic model is higher, on the contrary, if the p value is small, we believe that the performance of the basic model is worse than that of the comparison model examined. The SPA test results of different models are shown in Table 11, indicating that the prediction accuracy of the proposed model under the MSE and MAE loss function standards is optimal.

Table 11. SPA test results of different models.

		Comparison model			
Loss function	Base model	Proposed	LSTM	ERNN	GRU
MSE	Proposed	-	1.0000	1.0000	1.0000
	LSTM	0.0000	-	0.6420	0.5310
	ERNN	0.0000	0.3580	-	0.4150
	GRU	0.0000	0.4690	0.5850	-
MAE	Proposed	-	1.0000	1.0000	1.0000
	LSTM	0.0000	-	0.9950	0.5840
	ERNN	0.0000	0.0050	-	0.0170
	GRU	0.0000	0.4160	0.983	-

5. Empirical research of variational learning rate

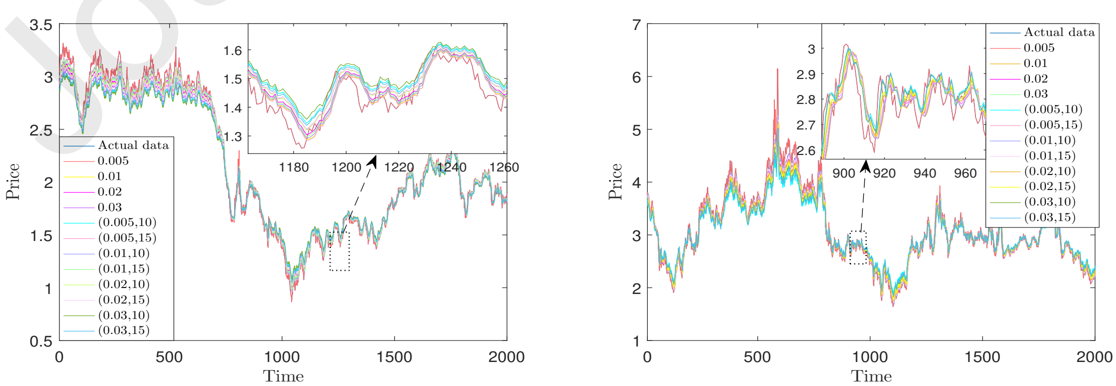
In Section 2.4, the variational learning rate is proposed. This arithmetic can help the model achieve better training results, avoiding poor prediction results due to inappropriate selection of learning rates, and reducing the possibility of falling into local optimal solutions. In this paper, the variational learning rate arithmetic is applied to the learning process of the ERNN model, and the prediction accuracy of the model is optimized. In the process of application, two parameters are selected: α and η_0. In order to better optimize the parameter selection, we have done empirical research on VR. The closing prices of Heating oil (HO) from Jan. 2012 to Sept. 2019 and Henry Hub natural gas (NG) from Oct. 2011 to Sept. 2019 are selected as the research data. Based on previous experimental analysis, we selected twelve sets of models with different parameters to compare the error metrics of the predicted results with HO and NG predictions, and stipulate an optimal parameter selection reference. 0.05, 0.01, 0.02, 0.03, $(0.005, 10)$, $(0.005, 15)$, $(0.01, 10)$, $(0.01, 15)$, $(0.02, 10)$, $(0.02, 15)$, $(0.03, 10)$, $(0.03, 15)$ represent twelve models of different parameters, respectively. The first four do not apply the VR using fixed learning rate as comparison models. Figs. 14(a)(b) show the prediction results for each model for NG and HO, Table 12 and Table 13 give the error metrics for each model, and Figs. 14(c)(d) depict the variation of the q-DSCID values for each model at different τ.

Table 12. Metrics of variational learning rate algorithm with different parameters for HO.

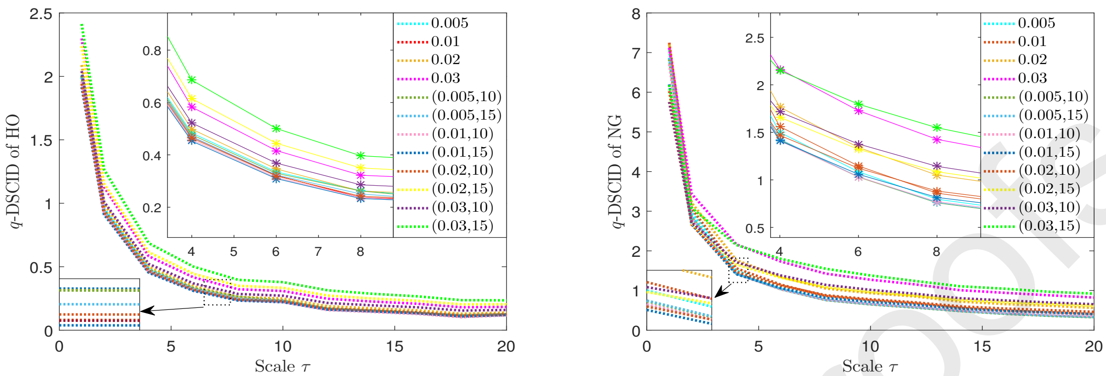
Model	Metrics					
	R^2	MAE	TIC	RMSE	SMAPE	q-DSCID
0.005	0.9105	0.0350	0.0114	0.0461	1.7471	2.0123
0.01	0.9212	0.0331	0.0107	0.0433	1.6574	2.0028
0.02	0.9222	0.0338	0.0107	0.0431	1.6859	2.0894
0.03	0.9112	0.0374	0.0115	0.0464	1.8479	2.2984
(0.005, 10)	0.9124	0.0346	0.0113	0.0456	1.7254	2.0004
(0.005, 15)	0.9169	0.0336	0.0110	0.0444	1.6799	1.9911
(0.01, 10)	0.9215	0.0328	0.0107	0.0432	1.6381	1.9606
(0.01, 15)	0.9303	0.0316	0.0101	0.0407	1.5777	1.9147
(0.02, 10)	0.9292	0.0321	0.0102	0.0411	1.6006	1.9419
(0.02, 15)	0.9076	0.0388	0.0118	0.0475	1.9141	2.2385
(0.03, 10)	0.9222	0.0347	0.0108	0.0433	1.7186	2.0412
(0.03, 15)	0.8949	0.0421	0.0127	0.0509	2.0708	2.4127

Table 13. Metrics of variational learning rate algorithm with different parameters for NG.

Model	Metrics					
	R^2	MAE	TIC	RMSE	SMAPE	q-DSCID
0.005	0.9150	0.0832	0.0232	0.1351	2.7478	7.0618
0.01	0.9198	0.0800	0.0226	0.1312	2.6263	7.2480
0.02	0.9140	0.0846	0.0234	0.1359	2.7922	7.2041
0.03	0.8903	0.0991	0.0264	0.1535	3.3195	7.1158
(0.005, 10)	0.9215	0.0793	0.0223	0.1298	2.6193	6.8311
(0.005, 15)	0.9217	0.0792	0.0223	0.1297	2.6156	6.8692
(0.01, 10)	0.9261	0.0762	0.0216	0.1260	2.5032	5.7033
(0.01, 15)	0.9388	0.0741	0.0203	0.1206	2.3478	5.1987
(0.02, 10)	0.9265	0.0766	0.0216	0.1256	2.5119	6.0270
(0.02, 15)	0.9148	0.0854	0.0232	0.1352	2.8399	6.7576
(0.03, 10)	0.9109	0.0879	0.0238	0.1383	2.9299	7.3544
(0.03, 15)	0.8800	0.1062	0.0276	0.1606	3.5905	7.1787



(a) (b)



(c) (d)

Fig. 14. (a)(b) Implementation results with variational learning rate under distinct parameters of HO and NG. (c)(d) Variation of q-DSCID values under different parameters with respect to the scale τ of HO and NG.

For a fixed α, it is found that the change in η_0 has little change in the performance of the model, even if the η_0 adjustment is large. When η_0 is fixed, α is fine-tuned, and it can be observed that the indicators of the model have obvious changes. The indexes of $(0.01, 15)$ are all optimal, and its q-DSCID value is always the smallest, indicating that the current parameters are helpful for improving the prediction accuracy of the model. Selecting the current parameters, after repeated experiments and training on different data sets (HO, NG, WTI, BRE, PRT, SNP), it is found that $(0.01, 15)$ greatly improved the training results of the model and is better than the optimal situation of a large number of rate experiments. Consequently, this work can draw empirical advice, when using the VR training model, it can give priority to the parameter selection $(0.01, 15)$ which encompass exceptional effect on a variety of data. Of course, it may get better training results with various values around $\alpha = 0.01$.

6. Conclusion

Implementing more accurate predictions of crude oil price volatility in the current situation of supply imbalance, instability of world economic market and frequent uncertainties is a difficult undertaking. This research constructs a novel multi-hybrid neural network prediction model, and achieves high-precision forecasting and analysis of crude oil futures prices.

The proposed model integrates EWT, deep bidirectional LSTMNN, ERNN with variational learning rate, random inheritance formula error correction algorithm. Also, step-by-step predictions are performed by selecting appropriate models according to the characteristics of the feature sequence. The multi-model comparison analysis indicates that the proposed model has more prominent performance, and the error metrics for BRE and WTI prediction results are optimal. The q-DSCID value of the proposed model is the smallest at different scales τ, and the ratio of performance improvement IP over LSTM is the largest. The results of comprehensive comparative analysis can be concluded that RIF error correction algorithm, deep bidirectional training structure and step-by-step predictions play an obvious role in model optimization. The experimental results of the variational learning rate illustrate that the proposed variational learning rate can optimize the model training process, ameliorate the prediction accuracy of the model.

In summary, the proposed EWT-RIF-DBLSTMNN-VRERNN model has good results in crude oil futures price forecasting, and the results obtained are more accurate and reliable, and can provide important references for relevant investors and national departments. It is prospected that the model can be further extended to other fields, better realize the application value and further optimize, and continuously enhance the prediction preciseness of the model.

Credit Author Statement

Bin Wang: Conceptualization; Methodology; Investigation; Software; Visualization; Writing - original draft and review & editing. **Jun Wang:** Conceptualization; Methodology; Investigation; Supervision; Writing - review & editing.

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Highlights

(1) Multi-hybrid neural network with EWT extraction and q-DSCID evaluation is proposed.

(2) Novel algorithm based on random inheritance formula is advanced and engrafted.

(3) Ameliorated DBLSTMNN, integrates deep bidirectional training configuration.

(4) Variational learning rate arithmetic is put forward to optimize parameter selection.

(5) Results show the high accuracy of the proposed model for crude oil futures.

Credit Author Statement

Bin Wang: Conceptualization; Methodology; Investigation; Software; Visualization; Writing - original draft and review & editing. **Jun Wang:** Conceptualization; Methodology; Investigation; Supervision; Writing - review & editing.

Declaration of interests

☒ The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

☐The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: